

3	(a)	is a value that is numerically equal to the thermal energy transferred when <u>unit mass</u> of a substance <u>changes state</u> , <u>without any change of temperature</u> .	
	(b)	(i) Straight line graph with <u>constant gradient</u> (Note: total mass M decreases linearly with time which means that mass of liquid evaporated increases linearly with time)	
		(ii) $Q = Pt = ml_v$ (where m is the mass that has evaporated) $P = \frac{m}{t} l_v$ $\frac{m}{t} = -\frac{\Delta M}{t}$ (Note: $\frac{\Delta M}{t}$ is a negative gradient of the graph in Fig 3.2 [where ΔM is the mass loss]) Since gradient of graph = $-\frac{\Delta M}{t}$ $P = (-\frac{\Delta M}{t})l_v$ or [power = - gradient $\times l_v$] Determine gradient of graph (or two points separated by at least 3.5 minutes) $120 = l_v \times \frac{(372 - 325) \times 10^{-3}}{(7.0 - 0) \times 60}$ $l_v = 1.07 \times 10^6 \text{ J kg}^{-1}$ (accept 2 s.f.) [will get a negative gradient but times negative will become positive]	
	(c)	Some energy/ heat is lost to surroundings so value is an overestimate	

		In calculating for l_v in part bii, heat lost was not taken into consideration $P = \left(-\frac{\Delta M}{t}\right)l_v$ However, if heat lost is taken into consideration, $P = \left(-\frac{\Delta M}{t}\right)l_v + Q$ l_v will be a smaller value. Therefore, l_v found in bii is an overestimation.	
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4	(a)	The sources must be coherent ; i.e. they must maintain a <i>constant phase difference</i> with respect to each other. OR The two wave sources must also emit waves of roughly the same amplitude.				
	(b)	(i)	1.	They are coherent as the phase difference between them is a constant value of π .		
			2.	From the graph, the waves arrive at point Y in antiphase, and interfere destructively.		
		(ii)	Path difference, $= QY - PY = \left(\sqrt{6^2+3^2} - \sqrt{6^2+1^2}\right) d$ $= 0.625 d$			
		(iii)	3 rd order minimum is formed at Y (can see from graph that the waves meet in antiphase, hence destructive interference occurs.) Path difference $= 0.625 d = 2.5 \lambda$ $\lambda = 0.25 d$ From $v = f \lambda$ $f = \frac{v}{\lambda}$ $f = \frac{3.0 \times 10^8}{0.25d}$ $= \frac{1.2 \times 10^9}{d} \text{ Hz}$			

5	(a)		The first order maximum is diffracted at an angle $\theta = \tan^{-1}(0.46/3.99) = 13.05^\circ$		
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